Math 310 **Fall 2026**
Quiz 5 Solutions **Drew Armstrong**

Problem 1. Compute the surface area of the following parametrized surface in $\mathbb{R}^3$:

$$\mathbf{r}(u,v) = (u, v, u+2v),$$

for $0 \leq u \leq 1$ and $0 \leq v \leq 1$.

The surface area is defined as

$$\int_0^1 \int_0^1 1 \, \|\mathbf{r}_u \times \mathbf{r}_v\| \, du \, dv.$$

We first compute the tangent vectors

$$\mathbf{r}_u(u,v) = \langle 1, 0, 1 \rangle,$$
$$\mathbf{r}_v(u,v) = \langle 0, 1, 2 \rangle,$$

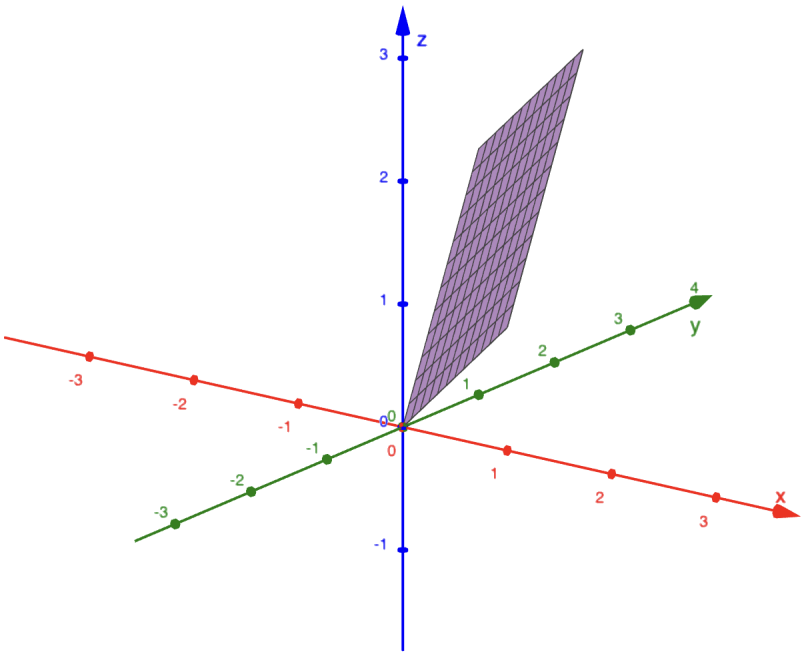
and then the cross product

$$\mathbf{r}_u \times \mathbf{r}_v = \langle -1, -2, 1 \rangle,$$
$$\|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{6}.$$

So the surface area is

$$\int_0^1 \int_0^1 1 \, \sqrt{6} \, du \, dv = \sqrt{6}.$$

In fact, this surface is just the parallelogram spanned by $\langle 1, 0, 1 \rangle$ and $\langle 0, 1, 2 \rangle$. Here is a picture:



Problem 2. Compute the integral of the vector field $\mathbf{F}(x, y) = \langle x, 1\rangle$ along the parametrized curve $\mathbf{r}(t) = (t^2, t^3)$ for $0 \leq t \leq 1$.

This integral is defined as

$$\int_0^1 \mathbf{F}(\mathbf{r}(t)) \bullet \mathbf{r}'(t)\, dt.$$

First we compute

$$\begin{aligned}
\mathbf{F}(\mathbf{r}(t)) &= \mathbf{F}(t^2, t^3) = \langle t^2, 1\rangle, \\
\mathbf{r}'(t) &= \langle 2t, 3t^2\rangle, \\
\mathbf{F}(\mathbf{r}(t)) \bullet \mathbf{r}'(t) &= \langle t^2, 1\rangle \bullet \langle 2t, 3t^2\rangle = 2t^3 + 3t^2.
\end{aligned}$$

Hence

$$\begin{aligned}
\int_0^1 \mathbf{F}(\mathbf{r}(t)) \bullet \mathbf{r}'(t)\, dt &= \int_0^1 (2t^3 + 3t^2)\, dt \\
&= \left[\frac{2}{4}t^4 + t^3\right]_0^1 \\
&= \frac{2}{4} + 1 \\
&= \frac{3}{2}.
\end{aligned}$$

Alternatively, we can use the fundamental theorem of line integrals. It turns out that $\mathbf{F} = \nabla f$, where $f(x, y) = \frac{1}{2}x^2 + y$. Hence

$$\int_0^1 \mathbf{F}(\mathbf{r}(t)) \bullet \mathbf{r}'(t)\, dt = f(\mathbf{r}(1)) - f(\mathbf{r}(0)) = f(1, 1) - f(0, 0) = \frac{3}{2} - 0 = \frac{3}{2}.$$